

1 Solutions of exercises in §1 (REINFORCE)

Exercise 1.1 (Rock-Paper-Scissors game). You play against an opponent who selects actions in an i.i.d. manner, according to a known distribution $\mu = (\mu_R, \mu_S, \mu_P)$.

- Model this as an MDP with $\mathcal{S} = \{\text{Play, Won, Lost}\}$. What is the optimal policy?
- Now assume the opponent instead follows a policy such that each time they play a move, the probability of repeating the same move decreases by one third, and their first action is drawn from μ . How would the model and optimal policy change?

Solution. *Part a)* We model the game as follows:

State space: $\mathcal{S} = \{\text{Play, Won, Lost}\}$.

Actions: $\mathcal{A}(\text{Play}) = \{R, P, S\}$, $\mathcal{A}(\text{Won, Lost}) = \{\text{Continue}\} = \{C\}$.

Rewards:

- $r(s = \text{Win}, a = C) = 1$,
- $r(s = \text{Lost}, a = C) = -1$,
- all others zero.

Transitions:

- $p(s' = \text{Won} | s = \text{Play}, a = R) = \mu_S$ (We play Rock and opponent plays Scissors),
- $p(s' = \text{Lost} | s = \text{Play}, a = R) = 1 - \mu_P$,
- $p(s' = \text{Won} | s = \text{Play}, a = S) = \mu_P$,
- $p(s' = \text{Lost} | s = \text{Play}, a = S) = 1 - \mu_R$,
- $p(s' = \text{Won} | s = \text{Play}, a = P) = \mu_R$,
- $p(s' = \text{Lost} | s = \text{Play}, a = P) = 1 - \mu_S$,
- $p(s' = s | s, a = C) = 1$,

Objective: $\mathbb{E}\{\sum_{t=1}^{\infty} \gamma^{t-1} r(s_t, a_t) | s_1 = \text{Play}\}$.

Optimal policy: Since the goal is to maximize the probability of winning, the optimal action is a^* , such that it corresponds to the next action in $\{R, P, S\}$ after $\text{argmax}_A \mu_A$ where $A \in \{R, P, S\}$.

Part b) Now we define state space as $\mathcal{S} = \{\text{Play, Won, Lost}\} \times (\mathbb{R}, \mathbb{R})$ where two real numbers denote μ_R and μ_S . Note that $\mu_P = 1 - \mu_R - \mu_S$. Transitions change as follows:

- $p(s' = (\text{Won}, \frac{2}{3}\mu_R, \mu_P - \frac{1}{6}\mu_R) | s = (\text{Play}, \mu_R, \mu_P), a = R) = \mu_S$,
- $p(s' = (\text{Lost}, \frac{2}{3}\mu_R, \mu_P - \frac{1}{6}\mu_R) | s = (\text{Play}, \mu_R, \mu_P), a = R) = 1 - \mu_S$,

and similarly for $a = S$ and $a = P$. The optimal policy now becomes state-dependent and for state $s = (\text{Play}, \mu_R, \mu_S)$ is given by the next action in $\{R, P, S\}$ after $\text{argmax}_A \mu_A$ where $A \in \{R, P, S\}$.

Exercise 1.2 Assume now that the opponent's distribution μ is **unknown**, but fixed.

- Propose a policy parametrization (use softmax with linear action preferences).
- Explain how to apply REINFORCE to learn an optimal policy and derive gradients.

Solution. *Part a)* We use a softmax policy with linear action preferences:

$$\pi_{\theta}(a|s) = \frac{\exp(\theta^T x(s, a))}{\sum_b \exp(\theta^T x(s, b))} \quad \text{if } s = \text{Play}, a \in \{R, P, S\}.$$

Since in states Won, Lost there is only one available action, no parametrization is needed

for those states. The simplest feature vector is a one-hot encoding:

$$x(s = \text{Play}, a) = \begin{bmatrix} 1\{a = R\} \\ 1\{a = S\} \\ 1\{a = P\} \end{bmatrix},$$

where $1(\cdot)$ is the indicator function. Note that, for this choice, $\theta \in \mathbb{R}^3$.

Part b) In REINFORCE, we first select the initial parameters $\theta^{(0)}$ arbitrarily.

We then repeatedly play the game (i.e., generate a trajectory/episode) under the policy induced by $\theta^{(k)}$. After finishing the episode, we update our parameters using:

$$\theta^{(k+1)} = \theta^{(k)} + \alpha_k \left(\sum_{t=1}^T \nabla_{\theta} \log \pi_{\theta}(a_{t,k} | s_{t,k}) \right) \left(\sum_{u=1}^T r(s_{u,k}, a_{u,k}) \right),$$

where $s_{t,k}$ is the state at time t in the current episode k . The expression for $\nabla_{\theta} \log \pi_{\theta}(a_{t,k} | s_{t,k})$ can be computed using simple derivative rules:

$$\begin{aligned} \nabla_{\theta} \log \pi_{\theta}(a | s) &= \nabla_{\theta} \log \frac{\exp(\theta^T x(s, a))}{\sum_b \exp(\theta^T x(s, b))} \\ &= \nabla_{\theta} (\theta^T x(s, a) - \log \sum_b \exp(\theta^T x(s, b))) \\ &= x(s, a) - \frac{\sum_b \exp(\theta^T x(s, b)) x(s, b)}{\sum_{b'} \exp(\theta^T x(s, b'))} \\ &= x(s, a) - \sum_b \pi_{\theta}(b | s) x(s, b). \end{aligned}$$

However, note that $\nabla_{\theta} \log \pi_{\theta}(a_{t,k} | s_{t,k}) \in \{\text{Won}, \text{Lost}\} = 0$.

Exercise 1.3 First show that $\sum_a \pi_{\theta}(a | s) \nabla_{\theta} \log \pi_{\theta}(a | s) = 0$. Then, using this equation and Markovian property, show the *causality principle*:

$$\mathbb{E}_{\tau \sim \pi_{\theta}} [\nabla_{\theta} \log \pi_{\theta}(a_t | s_t) r(s_u, a_u)] = 0 \quad \text{for } u < t.$$

Solution. For $u < t$, we have:

$$\begin{aligned} &\mathbb{E} [\nabla_{\theta} \log \pi_{\theta}(a_t | s_t) r(s_u, a_u)] \\ &= \sum_{s, a, s', a'} \mathbb{P}(s_t = s', a_t = a', s_u = s, a_u = a) r(s, a) \nabla_{\theta} \log \pi_{\theta}(a' | s') \\ &= \sum_{s, a, s', a'} \pi_{\theta}(a' | s') \mathbb{P}(s_t = s', s_u = s, a_u = a) r(s, a) \nabla_{\theta} \log \pi_{\theta}(a' | s') \end{aligned}$$

where the first equality follows from the Markovian assumptions on the system and the policy. Focusing only on terms containing a' we get:

$$\begin{aligned} \sum_{a'} \pi_{\theta}(a' | s') \nabla_{\theta} \log \pi_{\theta}(a' | s') &= \sum_{a'} \pi_{\theta}(a' | s') \frac{\nabla_{\theta} \pi_{\theta}(a' | s')}{\pi_{\theta}(a' | s')} = \sum_{a'} \nabla_{\theta} \pi_{\theta}(a' | s') \\ &= \nabla_{\theta} \sum_{a'} \pi_{\theta}(a' | s') = \nabla_{\theta} 1 = 0. \end{aligned}$$

and thus the whole term satisfies $\mathbb{E} [\nabla_{\theta} \log \pi_{\theta}(a_t | s_t) r(s_u, a_u)] = 0$.

2 Solutions of exercises in §2 (TRPO)

Exercise 2.1 Show that for every π, π' : $J(\pi') = J(\pi) + \mathbb{E}_{\tau \sim \pi'} \left[\sum_{t=1}^{\infty} \gamma^{t-1} A_{\pi}(s_t, a_t) \right]$.

Solution. Note that $A_{\pi}(s_t, a_t) = Q^{\pi}(s_t, a_t) - V^{\pi}(s_t) = r(s_t, a_t) + \gamma \mathbb{E}_{s_{t+1}} V^{\pi}(s_{t+1}) - V^{\pi}(s_t)$. Recall that the notation $\mathbb{E}_{\tau \sim \pi'}$ means that we take the expectation over $s_1 \sim \rho(s), a_1 \sim \pi'(\cdot|s_1), s_2 \sim P(\cdot|s_1, a_1), \dots$. Thus, it already contains the expectation over s_{t+1} , so we can write:

$$\begin{aligned} \mathbb{E}_{\tau \sim \pi'} \left[\sum_{t=1}^{\infty} \gamma^{t-1} A_{\pi}(s_t, a_t) \right] &= \mathbb{E}_{\tau \sim \pi'} \left[\sum_{t=1}^{\infty} \gamma^{t-1} (r(s_t, a_t) + \gamma V^{\pi}(s_{t+1}) - V^{\pi}(s_t)) \right] \\ &= \mathbb{E}_{\tau \sim \pi'} \left[\sum_{t=1}^{\infty} \gamma^{t-1} r(s_t, a_t) \right] + \mathbb{E}_{\tau \sim \pi'} \left[\sum_{t=1}^{\infty} (\gamma^t V^{\pi}(s_{t+1}) - \gamma^{t-1} V^{\pi}(s_t)) \right] \\ &= \mathbb{E}_{\tau \sim \pi'} \left[\sum_{t=1}^{\infty} \gamma^{t-1} r(s_t, a_t) \right] - \underbrace{\mathbb{E}_{\tau \sim \pi'} V^{\pi}(s_1)}_{= \mathbb{E}_{s_1 \sim \rho} V^{\pi}(s_1)} = J(\pi') - J(\pi) \end{aligned}$$

This completed the proof.

★ **Exercise 2.2** Let $C = \max_{s,a} |A_{\pi}(s, a)|$. Show the following two inequalities:

$$a) \max_s |\mathbb{E}_{a \sim \pi'(\cdot|s)} A_{\pi}(s, a)| \leq 2CD_{\text{TV}}^{\max}(\pi, \pi'), \quad b) \sum_s |d^{\pi}(s) - d^{\pi'}(s)| \leq \frac{2\gamma}{1-\gamma} D_{\text{TV}}^{\max}(\pi, \pi').$$

Solution. *Part a)* By definition, $A_{\pi}(s, a) = Q^{\pi}(s, a) - V^{\pi}(s)$, and since $V^{\pi}(s) = \sum_a \pi(a|s) Q^{\pi}(s, a)$, we have:

$$\mathbb{E}_{a \sim \pi(\cdot|s)} A_{\pi}(s, a) = \underbrace{\sum_a \pi(a|s) Q^{\pi}(s, a)}_{V^{\pi}(s)} - \underbrace{V^{\pi}(s)}_1 \sum_a \pi(a|s) = 0$$

Thus, subtracting this term from $\mathbb{E}_{a \sim \pi'(\cdot|s)} A_{\pi}(s, a)$ gives:

$$\begin{aligned} |\mathbb{E}_{a \sim \pi(\cdot|s)} A_{\pi}(s, a)| &= |\mathbb{E}_{a \sim \pi'(\cdot|s)} A_{\pi}(s, a) - \mathbb{E}_{a \sim \pi(\cdot|s)} A_{\pi}(s, a)| \\ &\leq \max_a |A_{\pi}(s, a)| \sum_a |\pi'(a|s) - \pi(a|s)| \end{aligned}$$

Taking maximum over states we recover inequality a).

Part b) Let's start from definition of d^{π} : $d^{\pi}(s) = (1-\gamma) \sum_{t=1}^{\infty} \gamma^{t-1} \mathbb{P}_{\pi}(s_t = s)$. We can also write this in vector form as: $d^{\pi} = (1-\gamma) \sum_{t=1}^{\infty} \gamma^{t-1} \rho P_{\pi}^{t-1}$. Now, using this form:

$$d^{\pi} - d^{\pi'} = (1-\gamma) \sum_{t=1}^{\infty} \gamma^{t-1} \rho (P_{\pi}^{t-1} - P_{\pi'}^{t-1})$$

Consider the term in parentheses and rewrite it as a telescoping sum:

$$P_{\pi}^{t-1} - P_{\pi'}^{t-1} = \sum_{k=0}^{t-2} P_{\pi}^k (P_{\pi} - P_{\pi'}) P_{\pi'}^{t-2-k}$$

Substituting this back into the sum we get:

$$\begin{aligned} \max_s |d^\pi(s) - d^{\pi'}(s)| &= \|d^\pi - d^{\pi'}\|_{\max} = \|(1 - \gamma) \sum_{t=1}^{\infty} \gamma^{t-1} \rho(P_\pi^{t-1} - P_{\pi'}^{t-1})\|_{\max} \\ &\leq (1 - \gamma) \sum_{t=1}^{\infty} \gamma^{t-1} \|\rho \sum_{k=0}^{t-2} P_\pi^k (P_\pi - P_{\pi'}) P_{\pi'}^{t-2-k}\|_{\max} \end{aligned}$$

where we used the triangle inequality. Consider a single term in the norm above:

$$\begin{aligned} \rho P_\pi^k (P_\pi - P_{\pi'}) P_{\pi'}^{t-2-k}(x) &= \sum_s (\rho P_\pi^k)(s) \sum_{s'} (P_\pi(s'|s) - P_{\pi'}(s'|s)) \underbrace{P_{\pi'}^{t-2-k}(x|s')}_{\leq 1} \\ &\leq \max_s \sum_{s'} |P_\pi(s'|s) - P_{\pi'}(s'|s)| \underbrace{\sum_s |(\rho P_\pi^k)(s)|}_{=1} \end{aligned}$$

where we applied Cauchy-Schwarz inequality. Substituting this back into the previous inequality yields:

$$\max_s |d^\pi(s) - d^{\pi'}(s)| \leq (1 - \gamma) \sum_{t=1}^{\infty} \gamma^{t-1} (t - 1) \max_s \sum_{s'} |P_\pi(s'|s) - P_{\pi'}(s'|s)|$$

Recall that $\sum_{t=1}^{\infty} \gamma^{t-1} (t - 1) = \frac{\gamma}{(1 - \gamma)^2}$. Finally

$$\begin{aligned} \sum_{s'} |P_\pi(s'|s) - P_{\pi'}(s'|s)| &= \sum_{s'} \left| \sum_a \pi(a|s) P(s'|s, a) - \pi'(a|s) P(s'|s, a) \right| \\ &= \sum_{s'} \sum_a |\pi(a|s) - \pi'(a|s)| \max_a P(s'|s, a) = 2D_{\text{TV}}(\pi, \pi'), \end{aligned}$$

and taking maximum over states s , we obtain the inequality we wanted.

Exercise 2.3 (Main result) Show that the two objectives are close with distance at most:

$$|J(\pi') - L_\pi(\pi')| \leq \frac{2C\gamma}{(1 - \gamma)^2} D_{\text{KL}}^{\max}(\pi, \pi')$$

Solution. Let's first recall that these functions are given by:

$$\begin{aligned} J(\pi') &= J(\pi) + \frac{1}{1 - \gamma} \mathbb{E}_{s \sim d^{\pi'}(s), a \sim \pi'(a|s)} A_\pi(s, a) \\ L_\pi(\pi') &= J(\pi) + \frac{1}{1 - \gamma} \mathbb{E}_{s \sim d^\pi(s), a \sim \pi'(a|s)} A_\pi(s, a) \end{aligned}$$

Thus, difference is given by:

$$|J(\pi') - L_\pi(\pi')| = \frac{1}{1 - \gamma} \left| \sum_{s, a} (d^{\pi'}(s) - d^\pi(s)) \pi'(a|s) A_\pi(s, a) \right|$$

Applying Cauchy-Schwarz inequality we get:

$$|J(\pi') - L_\pi(\pi')| \leq \frac{1}{1 - \gamma} \cdot \sum_s |d^{\pi'}(s) - d^\pi(s)| \cdot \max_s \left| \sum_a \pi'(a|s) A_\pi(s, a) \right|$$

Using the results from Exercise 2.2, we bound the second term with a), and the first term with b), obtaining:

$$\begin{aligned} |J(\pi') - L_\pi(\pi')| &\leq \frac{1}{1-\gamma} \cdot \frac{2\gamma}{1-\gamma} D_{\text{TV}}^{\max}(\pi, \pi') \cdot 2C D_{\text{TV}}^{\max}(\pi, \pi') \\ &= \frac{4\gamma C}{(1-\gamma)^2} [D_{\text{TV}}^{\max}(\pi, \pi')]^2 \end{aligned}$$

Applying Pinsker's inequality then yields the desired bound.